THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — SCIENCE (086) · MID-TERM / HALF-YEARLY PRACTICE PAPER

Session 2026-27 · School Examination Pattern

Class: 9	Subject: Science	Set 01	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology (Q.1–16), Section B is Chemistry (Q.17–27) and Section C is Physics (Q.28–39).
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of the options in such questions.
3. Section A carries 32 marks, Section B 25 marks and Section C 23 marks. Marks are printed against each question.
4. Draw neat, labelled diagrams wherever required. Use g = 10 m s^{-2} wherever needed.

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. Cells of the pancreas secrete large amounts of digestive enzymes (proteins). Which pair of cell organelles will be especially abundant in these cells? **1**

(a) Rough ER and Golgi apparatus (b) Smooth ER and lysosomes
(c) Plastids and vacuoles (d) Centrosome and chloroplasts

2. Dry raisins are soaked in pure water for some hours and they swell. This is because water: **1**

(a) moves out of the raisin by osmosis
(b) moves into the raisin through its semi-permeable membrane, since water is more concentrated outside
(c) moves into the raisin because sugar moves out by osmosis
(d) moves in both directions equally

3. Grass grows back quickly after it is mown. This is mainly due to the activity of: **1**

(a) apical meristem at the stem tip
(b) lateral meristem
(c) intercalary meristem at the base of leaves
(d) cambium of the root

4. Which of the following statements about cardiac muscle is INCORRECT? **1**

(a) Its action is involuntary.
(b) Its cells are cylindrical and branched.
(c) Its cells show light and dark striations.
(d) Its cells are long, unbranched and multinucleate.

5. A solar panel of area 4 m^2 receives sunlight of insolation 0.8 kW m^{-2} for 5 hours. The solar energy received by the panel is: **1**

(a) 3.2 kWh (b) 16 kWh (c) 4 kWh (d) 20 kWh

6. A single-celled organism has no membrane-bound nucleus or organelles but has a cell wall. It is placed in the kingdom: **1**

(a) Protista (b) Fungi (c) Monera (d) Plantae

7. Sugarcane is propagated by planting pieces of its stem. This method of reproduction is called: **1**

(a) sexual reproduction (b) vegetative propagation
(c) budding (d) spore formation

8. **Assertion (A):** Plasmolysis occurs when a plant cell is placed in a hypertonic solution. **1**
Reason (R): Water moves out of the cell by osmosis and the protoplast shrinks away from the cell wall.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Ozone depletion allows more ultraviolet radiation to reach the Earth's surface. **1**
Reason (R): Ozone is formed only in the troposphere, close to the ground.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Define budding. Name one organism that reproduces by budding. **2**

OR

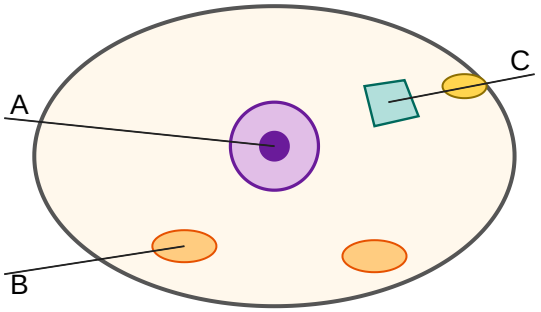
What is fragmentation? Give one example.

11. Name the type of tissue present in: (i) the hard shell of a walnut, (ii) the pulp of a ripe tomato, (iii) the lining of kidney tubules. **3**

OR

Name the muscle tissue found in: (i) the wall of the alimentary canal, (ii) the heart, (iii) the biceps attached to the bone. State one feature of each.

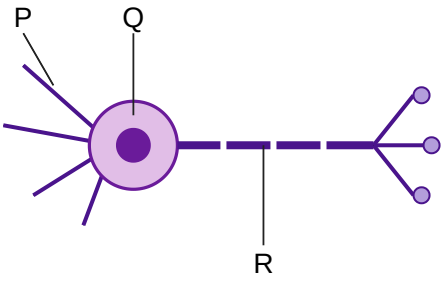
12. The figure shows an animal cell with three labelled parts A, B and C. Identify each and write one function of each. **3**



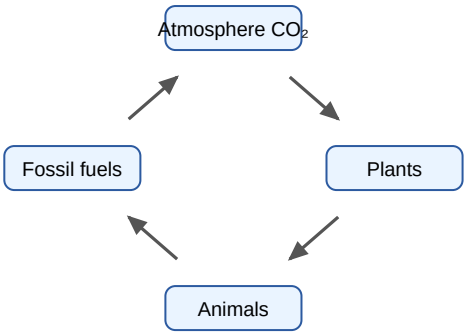
13. Copy and complete the table by writing 'Present' or 'Absent' for each cell: **3**

Structure	Human cheek cell	Spinach leaf cell	Bacterium
Cell wall			
Membrane-bound nucleus			
Chloroplast			

14. The figure shows a nerve cell (neuron) with parts P, Q and R labelled. Name P, Q and R and state in which direction the nerve impulse travels through the neuron. **3**



15. A forest research station studied how carbon moves in nature. Students drew the cycle shown below. They noted that the amount of CO_2 in the air has been rising since the burning of coal and petroleum increased, and that large areas of forest are being cleared for roads and farms.



(a) Name the process by which green plants remove CO_2 from the atmosphere. 1

(b) Name one human activity, other than deforestation, that increases atmospheric CO_2. 1

(c) Explain how deforestation can intensify the greenhouse effect. 2

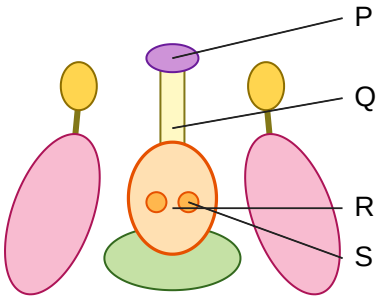
OR

(c) What is the greenhouse effect? Why is its enhancement harmful?

16. (i) What are lysosomes? Why are they called ‘suicide bags’ of the cell? **5**
(ii) State any three differences between prokaryotic and eukaryotic cells.

OR

Study the diagram of a flower. (i) Identify parts P, Q, R and S. (ii) Name the process by which pollen reaches the stigma. (iii) What do the ovule and the ovary develop into after fertilisation?



SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. A beam of light is passed through four liquids kept in transparent beakers. The path of the beam will be clearly visible in: **1**

(a) a salt solution (b) a glucose solution
(c) a starch solution (d) a copper sulphate solution

18. Whether water is obtained from a river, a well or by burning hydrogen in oxygen, hydrogen and oxygen are always found in the mass ratio 1 : 8. This illustrates the law of: **1**

(a) conservation of mass (b) constant proportions
(c) gaseous volumes (d) radioactive decay

19. An ion has 13 protons and 10 electrons. It is represented as: **1**

(a) Al^{3+} (b) Al^{+} (c) Na^{+} (d) Mg^{2+}

20. **Assertion (A):** Isotopes of chlorine, ^{35}Cl and ^{37}Cl, have identical chemical properties. **1**
Reason (R): Isotopes of an element have different numbers of protons.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

21. The thickness of a hair is 0.00008 m. Express it (i) in scientific notation and (ii) in micrometres (1 µm = 10^{-6} m). **2**

22. Name the technique used to separate (i) the different dyes present in black ink and (ii) cream from milk. State the principle behind each. **2**

23. Calculate the molecular mass of carbon dioxide (C = 12 u, O = 16 u) and the ratio by mass of carbon to oxygen in it. **2**

24. A saline solution is prepared by dissolving 18 g of sodium chloride in 282 g of water. (i) Calculate the mass percentage (% m/m) of the solution. (ii) How much sodium chloride is present in 500 g of this solution? **3**

25. State any three postulates of Bohr's model of the atom. **3**

OR

Differentiate between isotopes and isobars with one example of each.

26. A teacher tested three unlabelled samples P, Q and R, and recorded the observations in the table.

Sample	Appearance	Light path visible?	Settles on standing?	Residue on filter paper?
P	Clear	No	No	No
Q	Cloudy	Yes	Yes	Yes
R	Translucent	Yes	No	No

(a) Classify the samples P, Q and R as solution, suspension or colloid. 1

(b) Why can sample R not be separated by ordinary filtration even though it is not a true solution? 1

(c) Name the process to obtain the dissolved solute from sample P and write one example of sample P. 2

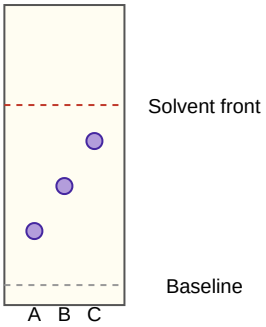
OR

(c) Arrange P, Q and R in increasing order of particle size and justify using the observations.

27. A mixture contains common salt, sand and iron filings. Describe, step by step, how you will separate the three components, naming the principle used at each step. **5**

OR

The chromatogram of a dye mixture is shown (solvent front is 100 mm above the baseline). (a) State the principle of paper chromatography. (b) Find the R_f value of spots A, B and C. (c) Which component is the most soluble in the solvent?



SECTION C — PHYSICS — 12 questions · 23 marks

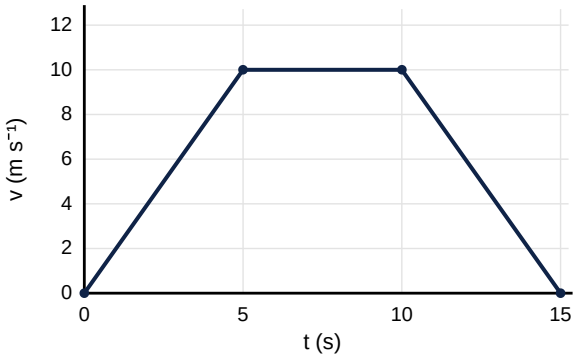
Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

28. A runner completes one full lap of a 400 m circular track in 80 s. Her average speed and average velocity for the lap are respectively: **1**

(a) 5 m s^{-1}, 0 (b) 0, 5 m s^{-1}

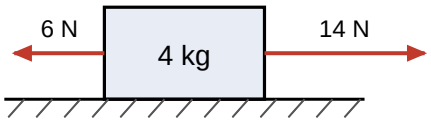
(c) 5 m s^{-1}, 5 m s^{-1} (d) 0, 0

29. The velocity–time graph of a car is shown. The distance covered by the car in the first 10 s is: **1**



(a) 50 m (b) 75 m (c) 100 m (d) 125 m

30. The figure shows the horizontal forces acting on a 4 kg block on a smooth surface. Its acceleration is: **1**



(a) 5 m s^{-2} (b) 2 m s^{-2} (c) 0.5 m s^{-2} (d) 8 m s^{-2}

31. A rocket moves upward by throwing hot gases downward. This is an example of Newton's: **1**

(a) first law (b) second law

(c) third law (d) law of gravitation

32. A 2 kg body is raised through a height of 5 m. The work done against gravity is (g = 10 m s^{-2}): **1**

(a) 10 J (b) 50 J (c) 100 J (d) 20 J

33. A sound wave has a frequency of 250 Hz and a wavelength of 1.36 m. The speed of the wave is: **1**

(a) 183.8 m s^{-1} (b) 1.36 m s^{-1} (c) 250 m s^{-1} (d) 340 m s^{-1}

34. **Assertion (A):** A force can change the direction of a moving body. **1**
Reason (R): A force can change the speed of a body.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

35. A car moving at 10 m s^{-1} accelerates uniformly at 2 m s^{-2} for 5 s. Find (i) its final velocity and (ii) the distance covered in this time. **2**

36. A cyclist stops pedalling but the bicycle keeps moving for some distance. Name the law that explains this and state why the cycle finally stops. **2**

OR

A gun recoils backward when a bullet is fired. Name the law that explains this and state why the recoil velocity is much smaller than the velocity of the bullet.

37. A pump raises 600 kg of water to a height of 20 m in 2 minutes. Calculate (i) the work done and (ii) the power of the pump (g = 10 m s^{-2}). **3**

38. During a hockey match, Seema, the goalkeeper, stops a ball of mass 0.16 kg moving at 25 m s^{-1}. The ball comes to rest in 0.05 s when she stops it with her padded gloves.

(a) Calculate the momentum of the ball before it is stopped. 1

(b) Find the average force exerted by the ball on her gloves. 1

(c) Why do goalkeepers pull their hands backward while stopping a fast ball? 2

OR

(c) If the ball were stopped in 0.2 s instead, what force would be exerted? Compare it with the first case.

39. A cyclist rides 6 km east in 20 minutes and then 8 km north in the next 20 minutes. Calculate (i) the total distance covered, (ii) the magnitude of displacement, (iii) the average speed and (iv) the average velocity. Why is the average speed greater than the average velocity? **5**

OR

A body starts from rest, accelerates uniformly at 2 m s^{-2} for 10 s, moves with constant velocity for the next 20 s and then comes to rest uniformly in 5 s. Find (i) the maximum velocity, (ii) the distance in the first 10 s, (iii) the distance at constant velocity, (iv) the retardation and (v) the total distance.